

Model Performance Assessment

Once you have a model which our learning algorithm has built using the training set, how can you say how good the model is? You use the test set to assess the model. The test set contains the examples that the learning algorithm has never seen before, so if our model performs well on predicting the labels of the examples from the test set, we say that our model generalizes well or, simply, that it's good.

To be more rigorous, machine learning specialists use various formal metrics and tools to assess the model performance. **For regression**, the assessment of the model is quite simple. A well-fitting regression model results in predicted values close to the observed data values. Let's go ahead and take a look at the most commonly-used metric scores to evaluate regression models.

Mean Absolute Error

The Mean Absolute Error gives us the average value of the total absolute differences between the predicted values output by the model and the actual values in the dataset. It is expressed in the same unit scale as the data measured, which makes it a straightforward metric to interpret. Values closer to 0 are considered better.

Let's observe the formula for the MAE score

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |Y_i - \hat{Y}_i|$$

Mean Squared Error

The formula for computing the Mean Square Error is very similar to the one we use to compute the Mean Absolute Error. This time, however, we square the differences between actual and predicted values for Y.

By squaring the differences, we penalize larger errors more than smaller errors, making it an ideal choice to evaluate models for tasks in which larger errors may lead to undesirable outcomes.

Let's observe the formula

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

Root Mean Squared Error

By using the Root Mean Squared Error we are able to bring the Mean Square Error back to the same unit scale as the observed data, which makes it more intuitive.

It still holds the same characteristics as the Mean Squared Error, penalizing larger errors more than smaller errors.

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2}$$

Coefficient of Determination (R^2)

The Coefficient of Determination - also referred to as R-Squared is a measure that tells us how well a regression model fits the actual data. It quantifies the degree to which the variance in the dependent variable is predictable from the independent variables. Values closer to 1.0 indicate a better model.

$$R^2 = 1 - \frac{\sum_{i=1}^n (Y_i - \hat{Y}_i)^2}{\sum_{i=1}^n (Y_i - \bar{Y}_i)^2}$$

In conclusion, the metric scores studied above are among the most commonly-used metrics to evaluate the performance of regression models.

For classification, things are a little bit more complicated. The most widely used metrics and tools to assess the classification model are:

Confusion Matrix

The confusion matrix in machine learning is the easiest way to measure the performance of a classification problem where the output can be of two or more type of classes. It is nothing but a table with two dimensions viz. "Actual" and "Predicted" and furthermore, both the dimensions have "True Positives (TP)", "True Negatives (TN)", "False Positives (FP)", "False Negatives (FN)" as shown below –

		Actual	
		1	0
Predicted	1	True Positives (TP)	False Positives (FP)
	0	False Negatives (FN)	True Negatives (TN)

Take an example of classifying emails as "spam" and "not spam" for a better understanding. Here a spam email is labeled as "positive" and a legitimate (not spam) email is labeled as negative.

Explanation of the terms associated with confusion matrix are as follows –

True Positives (TP) – It is the case when both actual class & predicted class of data point is 1. The classification model correctly predicts the positive class label for data sample. For example, a "spam" email is classified as "spam".

True Negatives (TN) – It is the case when both actual class & predicted class of data point is 0. The model correctly predicts the negative class label for data sample. For example, a "not spam" email is classified as "not spam".

False Positives (FP) – It is the case when actual class of data point is 0 & predicted class of data point is 1. The model incorrectly predicts the positive class label for data sample. For example, a "not spam" email is misclassified as "spam". It is known as a Type I error.

False Negatives (FN) – It is the case when actual class of data point is 1 & predicted class of data point is 0. The model incorrectly predicts the negative class label for data sample. For example, a "spam" email is misclassified as "not spam". It is also known as Type II error.

Classification Metrics Based on Confusion Matrix

1. Accuracy
2. Precision
3. Recall or Sensitivity
4. F1 Score
5. Type I Error Rate
6. Type II Error Rate

Accuracy

Accuracy is most common metrics to evaluate a classification model. It is ratio of total correction predictions and all predictions made. Mathematically we can use the following formula to calculate accuracy –

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN + TN}$$

Precision

Precision measures the proportion of true positive instances out of all predicted positive instances. It is calculated as ratio of the number of true positive instances and the sum of true positive and false positive instances.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall or Sensitivity

Recall (Sensitivity) is defined as the number of positives classifications by the classifier. We can calculate it with the help of following formula,

$$\text{Recall} = \frac{TP}{TP + FN}$$

F1 Score

F1 score is a balanced measure that takes into account both precision and recall. It is the harmonic mean of precision and recall. We can calculate F1 score with the help of following formula –

$$\text{F1 Score} = 2 \times \frac{(\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}}$$

Example: Let's take a practical example for a classifications of 10 emails as "spam" or "not spam". Here we are representing class for a spam email as positive (1) and a not spam email as negative (0).

The actual and predicted classes/ categories are as follows –

Actual Classification	0	1	0	1	1	0	0	1	1	1
Predicted Classification	0	1	0	1	0	1	0	0	1	1

Now find out the confusion matrix and some classification matrix as accuracy, precession, recall, F1 score from this example.

Model Selection Techniques in Machine Learning

1. Grid Search
2. Random Search
3. Bayesian Optimization
4. Cross-Validation Based Selection

Cross-Validation Based Selection

This method involves using cross-validation to evaluate multiple models and selecting the one with the best average performance. Instead of relying on a single train-test split, cross-validation divides the dataset into multiple parts and trains the model on different subsets. This helps to ensure that the model's performance is not just due to a specific split of data. By averaging the results from different splits, we get how well the model will perform on new, unseen data. This approach reduces the risk of overfitting and helps in selecting a good model.